

REPORT

Nereda Verdygo Package Plant Weert

Appendix 4.2: Control Narrative

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Version History

Version	Date	Key Changes
1	12/04/2024	First issue to the client.
1.1	22/04/2024	Correction phase vent
1.2	19/08/2024	Sludge buffer changed to Sludge holding tank. Update flushing of screens. Change to influent pump description.
1.3	24/09/2024	Added description for autosampler
1.4	06/12/2024	Changes to influent metal salt (2.2, 2.6) and to calamity pit (6.2, 6.6)
1.5	13/02/2025	Added logic for availability of blowers
1.6	01/05/2025	Updated logic sludge buffer valve
1.7	09/11/2025	Updated alarms, drives and effluent pump
1.8	04/12/2025	Added description for Aurea supply pump

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1 INTRODUCTION

1.1 General

The Control Narrative describes the process control of the Nereda® reactors and associated buffer(s) of the Nereda® Verdygo Package Plant (NVPP) Weert. It is based on the Piping & Instrumentation Diagrams (P&ID's) and the document "Control Philosophy", belonging to this project.

This document is a base document for the technical design of the process automation software for the Nereda® process. Before software development for the process automation is started, the document must be updated. Due to continuous optimization, based on latest insights and practical experience, updating of the document to the latest state of the Nereda® technology is necessary.

It is noted that, based on the functionality described in this document, the customer must elaborate this document into a technical design that is in line with his own process automation standards.

1.2 Nereda® Controller

Control of the NVPP Weert is realized with a PLC in combination with the Nereda® Controller. This document describes the functionality of the PLC software. The functionality of the Nereda Controller in general is described in document "Section 4 - Process automation and ICT" of the Basic Design.

In the document "Control Philosophy" (appendix S 4.1), the interaction and interfaces between Nereda® Controller and PLC are detailed. This document has to be read first before the document is processed into technical design. The software for the Nereda® Controller itself is supplied by RHDHV.

The hierarchical structure is depicted in the figure below:

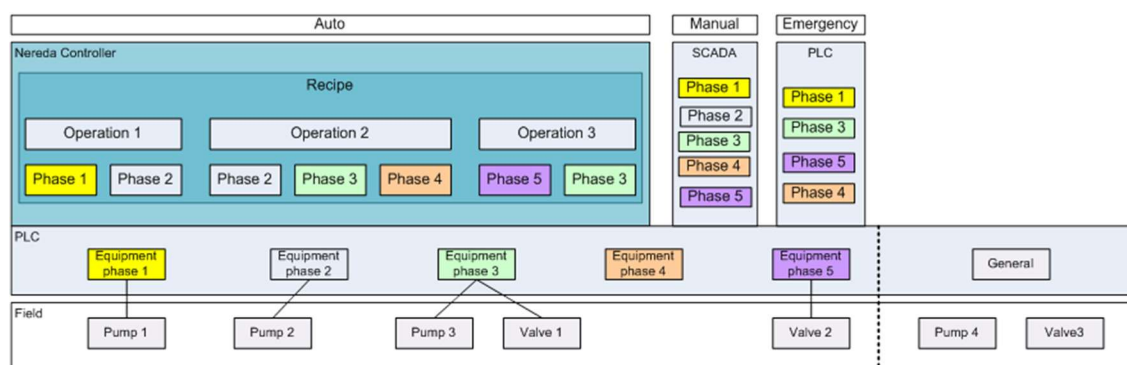


Figure 1 – Structure of Nereda® Controller, process automation and field

A complete (interface) list of data to be exchanged between Nereda® Controller and PLC will be made available in Excel format prior to the FAT.

1.3 Process parts

The installation consists of the following process components:

- Influent buffer
- Nereda® reactors
- Aeration system (e.g. blowers, valves)
- Sludge buffer
- Chemical dosing system

The Nereda® reactors and sludge buffer are batch processes. These processes are controlled using specific ISA-88 based batch control software.

1.4 General principles

In this paragraph some standard control principles are described which may be considered as a guideline for the elaboration of the technical design. These are commonly used basic functions in PLC-software applications. Slight deviations from this are possible, depending on the customer process automation standards.

1.4.1 Valves

The control of valves is incorporated in various sequences. For valves, the following items should be presented:

- Position open / closed
- Opening / closing (neither of limit switches active)
- Failure

Opening and closing of valves should be monitored (principle: ACOF: Automatic Check Of Feedback, safeguarding). The Open and Closed position readings should follow the Open and Close commands and the combination of the Opened and Closed readings should be valid. If a valve does not reach the position in which it is controlled, an alarm should be generated that properly describes the situation. The default action after an alarm occurs is Close (in case of 'fail safe' position).

1.4.2 Available status

An equipment has a status 'ready' or 'available' when it is switched via SCADA in Auto mode and it has no failures or alarms.

1.4.3 Shutdown at power failure

The Windows-based server on which the Nereda Controller runs, is powered via a smart UPS (with software control). In the event of a mains failure, this system ensures an uninterrupted supply voltage for at least 30 minutes, so that several software actions can automatically take place for a correct and controlled shutdown of the Windows-based server at low battery level of the UPS.

1.4.4 Phased restart after power failure

The process automation system should be equipped with a phased reset mechanism (after power failure). This implies that if the power comes back all phases must be reset to their rest situation.

The PLC will communicate if power is available through the PowerOK signal. If the power fails, this signal should become 'false' immediately. Only then, phase transitions should occur within the PLC based on the Operating Conditions (e.g. a valve which has lost power becomes unavailable and therefore the Discharge Phase which was in RUN will go to ABORTED through ABORTING). After the power returns the control system should wait an adjustable time (0- 5 minutes) before it automatically generates a general reset command. The general reset will cause all phases to go to the Ready or Interlocked state. Only after the general reset command the PowerOK signal can be made healthy. As soon as the PowerOK signal is true again the Nereda Controller can take over and will start the phases which are required to start.

The interaction between Nereda® Controller and process automation in case of power failure, is described in detail in document "Section 4 - Process automation and ICT" of the Basic Design.

1.4.5 Communication monitoring

Communication between the Nereda® Controller and PLC will be monitored by means of a watchdog signal.

The control system initiates the first watchdog signal, it sends a Boolean signal to the Nereda® Controller. From the Nereda® Controller, the process automation system will receive a Boolean signal of 0 or 1. The process automation system will check the value from the Nereda® Controller and send back the inverse bit (if Nereda sends 1, process automation system will send 0 and vice-versa). Every time the Nereda Server will send back the same value that the process automation system sent.

In case the process automation system will not receive the same value that it sent to the Nereda® Controller within a predefined adjustable 'watchdog' time, it will generate a communication failure. If the Nereda® process control is in "Auto" mode, the emergency recipe is automatically started for each reactor for which the emergency recipe is set to 'On'.

After watchdog communication is restored, the Nereda® process control mode must be switched back to "Auto" or "Manual" by the operator. This is not an automatic action from the process automation or Nereda® Controller.

See also the document "Control Philosophy" (appendix S 4.1).

1.4.6 Feedback monitoring

For each automatically or manually given command by the process automation (e.g. on/off, start/stop), it has to be checked whether this operating command is executed. The On / Off or Start / Stop readings should follow the commands, the combination should be valid.

If commands are not followed within an adjustable time, an alarm should be produced that gives an accurate description of the situation. The default action after an alarm occurs is Off or Stop. Also, the equipment which did not respond should become unavailable to phases which make use of this equipment.

Feedback monitoring is not listed separately on the specification sheets.

1.4.7 Measurement failure

Measurement failure is derived from:

- The digital input 'malfunction measurement'.
 - Monitoring of the 4-20 mA signal $<3.5 \text{ mA}$ and $> 21 \text{ mA}$.
 - Or a combination of the above.
- Each failure of a measurement must be reported separately.

For redundant measurements additional applies:

- On the SCADA, one of the measurements is selected for control, this is the preferred instrument.
- The other (reserve) measurement is used to check the preferred measurement. If the two measurement signals differ more than an adjustable difference percentage for an adjustable time, an alarm will be generated.
- If the preferred measurement has a malfunction, the automation automatically switches over to the other measurement. The reserve measurement becomes the preferred measurement.

1.4.8 Operation level hierarchy

On the SCADA, control is possible via a software switch Manual-0-Automatic the 'Automatic' control is determined by the automation of the PLC control system.

The control of the plant has three operating levels, automatic, manual and field:

- 1 Automatic: via SCADA and PLC. The automation system controls the equipment on base of measurements in the field and supervises the process.
- 2 Manual: via SCADA the operator can intervene in the automatic control. Starting and stopping of equipment is possible. The automation system is still supervising the process, i.e. safety and machine interlocking signals in the software will still be active.
- 3 Field: by means of a local switch, equipment can be started and stopped unconditionally.

The priority is: field control, manual, automatic, with field control having the highest priority and automatic the lowest priority. On the SCADA, selection of manual or automatic operation is possible via a software switch Manual-0-Automatic.

1.4.9 Alarm handling

Any failure must be processed in an alarm. Alarm handling has to take place by the acceptance and acknowledgment of alarms. This means that every alarm needs to be manually accepted and reset by an operator. When an alarm is accepted it may only be reset if the cause of the alarm is no longer present. When an alarm is reset, it must remain available to view in an alarm log.

Deviation alarm

A deviation alarm must be generated if the PLC controls a process value (e.g. flow, level) whose measured value deviates more from the setpoint than an adjustable maximum for an adjustable time.

Critical site alarms

At least following alarms must be implemented in the PLC/SCADA automation system of the plant. These alarms should be considered as critical site alarms.

- Power loss
- Communication failure
- Nereda® process control mode switched (see Process Control Philosophy document)
- No blowers available

1.4.10 Settings

Minimum flow rates, waiting times and so on, as specified in the control narrative must be adjustable via SCADA. For parameters that need to be adjustable only via the PLC, the text '[PLC]' is added.

Signals from pressure switches, level measurements (both discrete inputs and software derived values), flow switches, etc., which have the effect of an alarm, should have a standard adjustable input delay of 0-15 seconds.

1.4.11 Alternating and standby operation

Where pumps are specified for duty + standby operation, the control part must be implemented as follows. The logic in following paragraphs is also applicable for duty + standby (or assist) blowers.

Alternating operation

When a pump must be put in operation, the pump that has been out of operation for the longest time, is selected as duty pump. If several pumps are running and one pump should be out of operation, the pump must be selected that is already in operation for the longest time.

The control system will also rotate the duty pumps by monitoring the run hours of pumps. Pumps which have been taken out of order for any reason will not be selected by the PLC / control system. Selection of the duty and standby pumps will normally be a PLC function (AUTORUN) by means of a duty rotation program.

Standby operation

For pumps or other equipment that have standby operation, on SCADA a choice has to be made between pump 1, 2 or automatic alternating operation (duty rotation program in PLC).

In case of failure with pumps with duty+standby configuration, an alarm is generated, and the standby pump will be put in operation. Also, if the present selected pump is not available (anymore), the other pump will take over the function.

1.4.12 Cyclic 'on/off' or 'pulse/pause' operation

Based on the cyclic control principle, a device is put into operation during the specified 'on' (pulse) time and taken out of operation during the 'off' (pause) time. For each device, both times must be adjustable and set separately via the SCADA system.

A pulse/pause or on/off cycle should always begin with the 'on' (pulse) time. Once a device is put into 'AUTO' mode from the SCADA-system or a cabinet switch, the cycle must start again.

If a value "0" is filled in for the 'off' (pause) time, this should be processed as continuously on. If a value "0" is entered for the 'on' (pulse) time, this should be processed as continuously off. Both values "0" for 'on' and 'off' also result in continuous 'off'.

1.5 Explanation specification sheets

The treatment plant consists of several process parts. In the description of the process control, the following items are listed for each process part:

- Automatic control that includes multiple equipment associated with the process part.
- Equipment associated with the process part.
- Instrumentation associated with the process part.

Each part of the process is described in a separate chapter. Each chapter begins with a description of the process part. Then, the equipment and instrumentation associated with the process part are described on specification sheets.

The layout of the description of a process part and of the specification sheets is explained below.

It is noted that any customer process automation standards may complement the basic specification sheets included in this document.

1.5.1 Description of a process part

Description

General description of the operation of the process part.

Automation

Functional description of the control of the process part, as a single part and in relation to other process parts. These include sequences and controls that affect multiple machines and suchlike.

Settings

Operating options in SCADA for this process part (group commands).

Interlocks

Descriptions of process interlocks, i.e. situations and signals that affect the normal operation. The signals mentioned here switch off the process part, lock it until the relevant signal is dropped and the process part is reset, and generate an alarm. If necessary, differences in manual mode and automatic mode will be indicated.

Presentation

Description of signals and / or events, to be presented for this process part, provided they do not belong to a measurement or machine.

1.5.2 Equipment specification sheet

Settings

SCADA

Operating options in SCADA for this equipment. Also, controller settings are listed here.

Automation

The normal operation of the equipment. In automatic mode, the machine must be switched on or off based on the conditions specified in the specification sheet. This also includes duty cycle / standby algorithms and suchlike.

In Manual mode the machine will operate continuously. Exceptions to this rule are described here.

Locking means that operation of equipment is blocked until the locking conditions are no longer present. After this, the equipment can come back in operation.

Interlocks

Description of the situations and signals that interfere with the Manual and Auto operation mode in normal operation. The signals mentioned here switch off the equipment, lock it until the relevant signal is dropped and locking is reset, and generate an alarm. If necessary, differences in manual mode and automatic mode will be indicated.

Presentation

Description of the presentation of operating status, switch settings and suchlike.

1.5.3 Instrumentation specification sheet

Presentation

Signals to be presented locally or centrally.

Switching and alarm points

The following items are presented in table form:

- Switching points:
the various switching points of a measurement. These can be hardware contacts or software derived contacts from the measurement signal.
- Action:
the actions to be executed when a switching point is triggered
- Adjustable (Ad):
indicating that the switching or alarm point needs to be adjustable on the SCADA
- Preference value (Pv):
default value of the switching or alarm point.
- Alarm (Al):
indication that the switch has an alarm function.
- Alarm text:
alarm text that needs to be mentioned in the alarm list.

A specification table looks like this:

SCADA

■ flow [m³/h]

Switch points	Action	Ad	Pv	Al	Alarm text
LL	Feed pump off	x		x	Flow too low (time delayed)
L		x		x	Flow low
Failure measurement				x	Failure flow measurement

Ad = adjustable; Pv = Preference Value; Al = Alarm.

2 INFLUENT FEED

2.1 Introduction

The influent buffer provides the supply to the Nereda® reactor during the feeding of the Nereda® reactor. In the container 1 frequency-controlled pump is installed.

In automatic mode, the desired flow rate is determined by the Nereda® Controller. Within the Nereda® Controller a prediction algorithm is used to predict the influent flow and to keep the optimal level in the influent buffer. The Nereda® controller determines the flow rate setpoint based on the level in the influent buffer and passes this to the PLC.

The influent in the influent buffer tank is continuously mixed with a mixer. Mixing is stopped if the level in the influent buffer is lower than an adjustable level.

2.2 Flow to Nereda® influent buffer

Influent pump 46 2 AP ifp 95 001

The flow from the upstream pumping pit (46 2 AP ifp 95 001) towards the Nereda® influent buffer should be based on an inversed proportional relationship with the influent buffer level, using flow measurement 46 2 VB FT 95 001 and level measurement MOD NVPP 001 LT 001. There should be two level setpoints in the influent buffer for the starting and stopping of the pump in the pumping pit. The feed pump cannot be started while the Grit screens is being flushed.

When the feed pump has stopped, drain valve 46 2 AP a—95 003 must be opened for an adjustable time to drain the pipe.

Influent frost protection

In freezing weather (-5°C) the influent pump (46 2 AP ifp 95 001) should be operated intermittently, preventing it from being out of operation for too long, even when the level in the influent buffer is above the H-level setpoint. The goal of this control is to prevent frost in the pipe upstream of the pumping pit. In this extreme case, it is acceptable to use the overflow of the influent buffer. The H-level setpoint should not stop the influent pump. The influent pump should be started if an adjustable time has passed since the influent pump was last in operation. After an adjustable duration the pump should stop.

Grit screens

The grit screening installation must be started when feed pump 46 2 AP ifp 95 001 is started and stopped and adjustable time after the feed pump has been stopped.

The grit screening installation contains two safety protections that must stop the grit installation. The first is a safety stop, if the safety stop is pressed, the installation must immediately stop and can only be restarted with after a reset command on the HMI. The second is a detection for the opening of the hatch. If the hatch is opened, the installation must stop. The installation can continue when the catch is closed.

Automation

Reactor feed flowcontrol

The feed pump must be started when a 'feed' phase of a Nereda® reactor is in RUN. To control the influent feed flow to the reactor, the feed pump must be PI-controlled on the flow setpoint from the Nereda® Controller.

Flow measurement failure

In case of flow meter failure, corrective actions must be taken in the PLC. The Nereda® Controller will still issue a flow setpoint to the PLC. In the PLC, the flow setpoint must be used to determine the number of pumps to be operated and the pump speed by using the theoretical pump capacity.

Interlocking

The flow control of the influent feed must be started when the equipment phase "Feed" of the Nereda® reactor is in the state RUN. When the feed phase is no longer active, the pump(s) must be stopped.

During the "RUN" state of the feed phase, the feed to the reactor must be paused in either of the following situations:

- Low level influent buffer.

2.2.1 Phase metal salt dosing

The metal salt dosing phase is used to dose metal salts into the influent buffer.

Parameters

- None

Start logic

- None

RUN logic

- Metal salt dosing flow must be controlled at a fixed flow for the duration of the RUN phase.

Stop logic

- None

Operating conditions

1. Reserved, do not use
2. No error from the metal salt dosing unit
3. No LL-level in the metal salt IBC 46 2 AP LT 95 002
4. Level measurement 46 2 AP LT 95 002 available
5. Spare
6. Spare
7. Spare
8. Spare
9. Spare
10. Spare
11. Spare
12. Spare
13. Spare

14. Spare

15. Spare

16. Spare

2.3 Settings

- Settings for flow control.
- Settings for on/off switching of pump and mixer.

2.4 Interlocking

- None

2.5 Presentation

- None

2.6 Specification sheets

2.6.1 Influent feed pump MOD NVPP 001 ctp 001 / 46 2 AP ifp 95 001

Settings

SCADA

- Manual-0-Auto
- Start/stop
- Capacity setpoint (manual)
- Controller settings (gain, integration time)
- Settings Low level for dry run protection

Automation

Automatic

In automatic mode, the pump is controlled as described in section 2.2.

Manual

In manual mode, the operator can start and stop the pump. The pump must be operated at a capacity setpoint, adjustable by the operator.

Interlocks

- Thermal overload motor
- Electric circuit failure
- Safety switch off
- Frequency inverter failure

- LL-flow (time delayed)

Presentation

- State (automatic/ manual/ field control/ safety switch)
- Started (read back start is true)/
- Stopped (read back stop is true)
- Failure
- Actual pump capacity

2.6.2 Mixer MOD NVPP 001 mng 001

Settings

SCADA

- Manual-0-Auto
- Start/Stop
- settings Low level for dry run protection

Automation

Automatic

In automatic mode, the mixer is operated continuously. The mixer is stopped if the actual influent buffer level is lower than an adjustable Low level.

Manual

In manual mode, the operator can start and stop the mixer.

Interlocks

- Thermal overload motor
- Moisture detection pump (MS) (no machine stop)
- Electric circuit failure
- Safety switch off

Presentation

- State (automatic/ manual/ field control/ safety switch)
- Started (read back start is true)/
- Stopped (read back stop is true)
- Failure

2.6.3 Level measurement influent buffer MOD NVPP 001 It 001

SCADA

- Level [m]

Switch points	Action	Ad	Pv	Al	Alarm text
H	Stop influent pump 46 AP ifp 95 001*			x	Level too High influent buffer
L1	Stop mixer	x			Level Low influent buffer
L2	Stop influent feed pump	x		x	Level too Low influent buffer
Failure measurement	Predictive control disabled by Nereda Controller			x	Failure level measurement influent buffer

Ad = adjustable; Pv = Preference Value; Al = Alarm.

*) Unless influent frost protection is active. See 2.2.

2.6.4 Level measurement metal salt IBC 46 2 AP LT 95 002

SCADA

■ Level [m]

Switch points	Action	Ad	Pv	Al	Alarm text
H		x		x	Level High metal salt
L		x		x	Level Low metal salt
LL	Stop metal salt dosing pump	x		x	Level too Low metal salt
Failure measurement				x	Failure level measurement metal salt IBC

Ad = adjustable; Pv = Preference Value; Al = Alarm.

2.6.5 Flow measurement reactor feed MOD NVPP 001 ft 001/ 46 2 VB FT 95 001

SCADA

■ Flow [m³/h]

Switch points	Action	Ad	Pv	Al	Alarm text
HH		x		x	Flow too high
L		x		x	Flow low
LL	Stop influent feed pump	x		x	Flow too low (time delayed)
Deviation alarm		x		x	Deviation alarm feed flow
Failure measurement	determine the number of pumps and pump speed by using the theoretical pump capacity, see 2.2			x	Failure flow measurement

Ad = adjustable; Pv = Preference Value; Al = Alarm.

3 NEREDA® REACTOR

3.1 Introduction

From the influent buffer the wastewater is pumped batch wise to the Nereda® reactors. The operation of the Nereda® reactors is controlled by a PLC and the Nereda® Controller.

If the communication between the Nereda® controller and the PLC should fail, it switches to an adjustable Emergency Recipe that is programmed into the PLC. See Chapter 7 and the description in the document “Control Philosophy” (appendix S 4.1).

3.2 Automation

The processes in the Influent buffer, Nereda® reactors and Sludge buffer are controlled by the Nereda® Controller. This control structure is modular. There are modules for the following processes:

Influent buffer:

- Metal salt dosing

Nereda® reactor:

- Feed (effluent discharge)
- Aerate
- Wait (settle)
- Sludge discharge
- Vent

Sludge buffer:

- Feed
- Sludge discharge

Each module in the Nereda® Controller has a PLC counterpart, the so-called equipment phases. The Nereda® Controller determines which equipment phase starts and stops. Also, additional parameters can be passed from the Nereda® Controller to the PLC such as a desired flow rate or a desired capacity. In the PLC, subsequently the corresponding equipment phase is executed. These are described in the following sections, for each equipment phase following is described:

- Phase parameters
- Actions during start of the phase (start logic)
- Actions during operation (run logic)
- Actions during stopping of the phase (stop logic)
- Operating conditions

All equipment phases are executed according to a fixed state diagram. The state diagram is shown below and is detailed in the document “Control Philosophy” (appendix S 4.1).

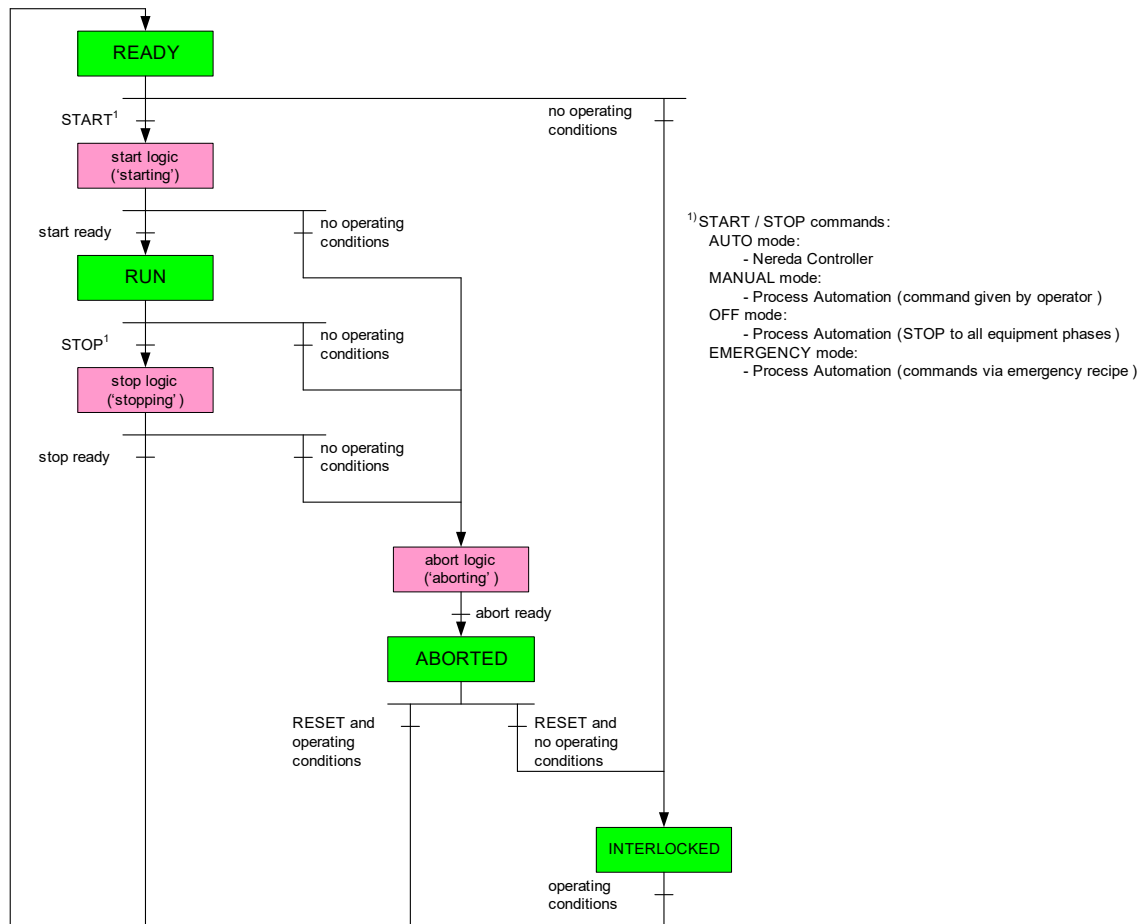


Figure 2 – State diagram for equipment phases

A programming example for the state diagram of an equipment phase in the PLC is included in Appendix A.

The status of the operating conditions of each phase is communicated with the Nereda® Controller, by means of a 16-bit status word. In the status word, there are fixed bits for the conditions, indicated by the number of the operating condition. If a condition is added in the future or during FAT, it will be added to the next available number. If a condition expires, it becomes an empty skipped position. The numbering of the conditions is not adjusted accordingly. The above is shown in the following table.

Table 1 — Example of operating conditions

Operating condition nr. (= bitnr. Of status word)	Operating Condition description	
1	Feed valve AV41101 is available	<i>Example condition</i>
2	Sludge discharge valve AV41103 is available OR closed	<i>Example condition</i>
3	Operating condition 3	
4	Operating condition 4	<i>Expired condition</i>
5	Operating condition 5 .	<i>Added condition</i>
..	...	
..	...	
	Spare	

16		
----	--	--

In the example of **Table 1** applies:

- OC_statusword = 0 if all Operating Conditions are true
- OC_statusword = 2 if Sludge discharge valve AV41103 is unavailable AND not closed
- OC_statusword = 3 if Sludge discharge valve AV41103 and Feed valve AV41101 are both unavailable

In the following paragraphs the phases, valves and instruments of 1 reactor are described. The phases for the other reactors must be realized in the same way.

3.2.1 Phase feed

In this phase, the reactor is filled via the influent feed pump. The control of the pump has been described under the process section “Influent feed” and should not be programmed as part of the phase, but as external control logic which is triggered if a feed phase is in RUN. The current paragraph is only the description for the starting and stopping of the feed phase. The feed phase controls feed valve MOD NVPP 001 a—007.

Parameters

- Flow ‘feed’ setpoint [m³/h]

Start logic

- Open feed valve MOD NVPP 001 a—007

RUN logic

- The pump will be operated on the flow setpoint parameter from the Nereda® Controller as described in paragraph 2.2

Stop logic

- Close feed valve MOD NVPP 001 a—007

Operating conditions

1. Feed valve MOD NVPP 001 a—007 is available
2. Sludge discharge valve MOD NVPP 001 a—023 is available OR closed
3. *Reserved, do not use*
4. *Reserved, do not use*
5. No HH-level (feed) in the reactor MOD NVPP 001 It 004
6. Level measurement MOD NVPP 001 It 004 available
7. Air valve influent grid MOD NVPP 001 a--067 is closed
8. Spare
9. Spare
10. Spare

11. Spare

12. Spare

13. Spare

14. Spare

15. Spare

16. Spare

3.2.2 Phase aerate

During the aeration phase, air is brought into the reactor. The desired capacity (0-100%) is specified by means of a parameter from the Nereda® Controller. In the PLC this capacity must be translated into activation and control of the aeration system. The aeration system consists of a combination of blowers and valves.

The selection, switching and control of the blowers and valves, is described in the “blower section” in chapter 4 and should not be programmed as part of the phase, but as external control logic which is triggered if a Aeration Phase of a reactor is in RUN.

Aeration via aeration grid or influent grid

During the aerate phase the reactor can be mixed by opening the air supply to the influent grid and closing the air supply to the aeration grid. This is indicated by the Nereda controller by separate parameters from the Nereda® Controller. The valves of the aeration and influent grid must be opened if the relevant parameter is active [=1] and closed if the parameter is not active [=0]

Parameters

- Aeration capacity setpoint [0-100%].
- Air valve aeration grid MOD NVPP 001 a--036 [0/1]
- Air valve influent grid MOD NVPP 001 a--067 [0/1]

Start logic

- Open valve MOD NVPP 001 a--036 AND/OR MOD NVPP 001 a--067 (depending on parameters)

RUN logic

- Control blower(s) with capacity setpoint from the Nereda® Controller. During the RUN valves MOD NVPP 001 a—036/-67 should be adjusted if the parameter changes.

Stop logic

- Close valve MOD NVPP 001 a--036 AND/OR MOD NVPP 001 a—067 (depending on parameters)

Operating conditions

1. *Reserved, do not use*
2. *Reserved, do not use*
3. *Reserved, do not use*
4. *Reserved, do not use*

5. *Reserved, do not use*
6. *Reserved, do not use*
7. *Reserved, do not use*
8. NOT (High level (aeration) in reactor MOD NVPP 001 It 004 and phase 'aerate' NOT ('starting' or 'RUN' or 'stopping'))
9. Level measurement MOD NVPP 001 It 004 available.
10. Air valve aeration grid MOD NVPP 001 a--036 is available.
11. Air valve influent grid MOD NVPP 001 a--067 is available.
12. spare
13. spare
14. spare
15. spare
16. spare

3.2.3 Phase wait

Parameters

- None

Start logic

- None

RUN logic

- None

Stop logic

- None

Operating conditions

1. spare
2. spare
3. spare
4. spare
5. spare
6. spare
7. spare
8. spare
9. spare

10.spare

11.spare

12.spare

13.spare

14.spare

15.spare

16.spare

3.2.4 Phase sludge discharge

The sludge discharge phase controls discharge valve MOD NVPP 001 a-- 023.

The amount of discharged sludge is monitored by measuring the level difference in the reactor during the state 'RUN'. If the level difference is lower than an adjustable value, an alarm is generated.

Parameters

- None

Start logic

- Open discharge valve MOD NVPP 001 a-- 023.

RUN logic

- None

Stop logic

- Close discharge valve MOD NVPP 001 a-- 023.

Operating conditions

1. Discharge valve NVPP 001 a-- 023 is available and valves for other reactors are closed
2. Reserved, do not use
3. Reserved, do not use
4. Reserved, do not use
5. Reserved, do not use
6. No HH-level in sludge buffer
7. spare
8. spare
9. spare
- 10.spare
- 11.spare
- 12.spare

13.spare

14.spare

15.spare

16.spare

3.2.5 Phase vent

The vent phase controls the venting of the influent grid and sludge grid. Which grid is vented is determined by the Nereda Controller via separate parameters.

The valves associated with the grid to be vented are controlled via the process automation, as followed:

Air grid: vent valve MOD NVPP 001 a—038 open, air pressure valve air grid MOD NVPP 001 a—036 open
 Influent grid: vent valve MOD NVPP 001 a—038 open, air pressure valve influent grid MOD NVPP 001 a—067 open

Parameters

- Venting air grid [0/1]
- Venting influent grid [0/1]

Start logic

- None

RUN logic

- Opening and closing the relevant valves for venting, based on the actual parameter status.

Stop logic

- Close vent valves MOD NVPP 001 a—038 and MOD NVPP 001 a—036
- Close air pressure valves MOD NVPP 001 a—036 and MOD NVPP 001 a—067

Operating conditions

1. Reserved, do not use
2. Vent valve MOD NVPP 001 A—038 is available
3. Reserved, do not use
4. Reserved, do not use
5. Air pressure valve MOD NVPP 001 A—036 is available
6. Air pressure valve MOD NVPP 001 A—067 is available
7. Spare
8. Spare
9. Spare

10. Spare

11. Spare

12. Spare

13. Spare

14. Spare

15. Spare

16. Spare

3.2.6 Settings

Adjustable values, as mentioned in the automation section above, must be made adjustable on SCADA.

3.2.7 Alarms

The status "Aborted" of each phase must be configured as an alarm.

3.2.8 Presentation

Presentation via the web pages of the Nereda® Controller and as indicated in the document "Control Philosophy" (appendix S 4.1).

3.3 Specification sheets

3.3.1 Valves reactor MOD NVPP 001 a--007/--023/--36/--38/--067

Settings

SCADA

- Manual-0-Auto
- Open / Close

Automation

Automatic

In automatic mode, the valves must be operated as described in the various equipment phases under chapter 3.2.

Manual

In manual control the operator can issue the following commands:

- ☐ Open
- ☐ Close

Interlocks

- Check on feedback failure (valve opening / closing)
- Electric circuit failure

Presentation

- State (automatic/ manual/ field control)
- Opened (read back open is true)
- Closed (read back close is true)
- Failure

3.3.2 Temperature measurement reactor MOD NVPP 001 41QT1002

SCADA

- Temperature [°C]

Switch points	Action	Ad	Pv	Al	Alarm text
H	-	x		x	Temperature reactor high
Failure measurement				x	Failure temperature measurement reactor

Ad = adjustable; Pv = Preference Value; Al = Alarm.

3.3.3 Oxygen measurement reactor MOD NVPP 001 41QT1003

SCADA

- Oxygen concentration [mg/l]

Switch points	Action	Ad	Pv	Al	Alarm text
Failure measurement	Action is taken by Nereda Controller, no action from PLC required. See "Section 4 - Process automation and ICT"			x	Failure oxygen measurement reactor

Ad = adjustable; Pv = Preference Value; Al = Alarm

3.3.4 Level measurement reactor MOD NVPP 001 It 004

SCADA

- Level [m]

Switch points	Action	Ad	Pv	Al	Alarm text
HH	Abort / interlock Feed phase -> operation conditions not met	x		X	Level too High to feed
H	Interlock Aerate phase reactor-> operation conditions not met	x		X	Level too High for aeration
Level L2 Sludge discharge	Stop Sludge discharge phase reactor during Emergency mode	x			

Level difference sludge discharge	Monitoring the amount of discharged sludge during phase 'sludge' discharge. See 3.2.4	x		x	Insufficient sludge discharge
LL	High priority alarm (reactor level too low, possible leakage)	x		x	Reactor level too low, "leakage"
Failure measurement	Abort / interlock Feed, Aerate phase -> operation conditions not met			x	Failure level measurement reactor

Ad = adjustable; Pv = Preference Value; Al = Alarm.

3.3.5 pH measurement reactor MOD NVPP 001 41QT1010

SCADA

■ pH [-]

Switch points	Action	Ad	Pv	Al	Alarm text
L		x		x	pH too Low reactor
H		x		x	pH too High reactor
Failure measurement				x	Failure pH measurement reactor

Ad = adjustable; Pv = Preference Value; Al = Alarm.

4 BLOWER SECTION

4.1 Introduction

The blower section controls the blowers and valves to provide the reactor with sufficient air. For the aeration of the Nereda® reactors, 2 frequency-controlled blowers are installed per reactor. The blower configuration is 1 duty + 1 assist per reactor.

Note that the PLC-system must ensure that the blowers and valves are switched correctly during states 'starting' and 'RUN' of the aeration phase. It is possible that 2 aerated phases can occur one after the other in a cycle, the statuses must then be able to switch from RUN-STOPPING-READY to STARTING-RUN again without any problems.

4.2 Automation

The blower system includes the following components:

- Blowers MOD NVPP 001 bl 001... MOD NVPP 001 bl 006

The aeration phase of each reactor provides a START command, STOP command and an air capacity setpoint for the blower section.

*Please note that this is **not** the same type of control as a frequently applied system based on oxygen demand (feedback control principle). The Nereda system determines the air demand based on different control modes and various parameters such as the NH_4 , NO_3 , Redox and O_2 -measurements in the reactors.*

Each reactor will demand a maximum of two blowers. The number of requested blowers (1 or 2) is determined by means of the capacity setpoint from the Nereda® Controller. The setpoint from the Nereda® Controller is in the range 0..100%. The blower capacity has to be scaled to the requested aeration capacity from the Nereda controller. On SCADA these settings have to be adjustable.

- 0% demand from the Nereda® Controller means 0 Nm^3/h .
- 100% demand from the Nereda® Controller means two blowers at maximum capacity.

Following table shows an example for the aerating capacity at peak air demand during DWF conditions.

Table 2 – Example scaling air capacity - blowercapacity

Blower capacity	Air capacity setpoint Nereda® Controller	Air flow reactor
1 blower 0 – 100 %	0 – 57%	114 Nm^3/h (max air capacity 1 blower)
2 blowers 0 – 86%	57% – 100%	197 Nm^3/h (max. air demand reactor)

A reactor that sends a start command must obtain at least one blower.

4.3 Blower control

The blowers are started when the Aeration Phase of a reactor is in RUN. The number of blowers in operation on the header will be primary determined by means of the requested air capacity by the Aeration phase in

STARTING/RUN, see section 3.2.2. In this way extra delay due to switching on/speeding up blowers one by one after each other, is avoided.

As soon as the aeration of a reactor starts, the total air demand will increase by a certain percentage. Based on this total air demand and the blower capacity ranges it must be determined how many blowers should be in operation to be able to meet the desired total air demand.

A typical procedure for switching blowers on/off to the header system is described below. The procedure may be changed in respect to the blower's technical specifications.

Blower switching on

IF

- The requested total air capacity is higher than the total maximum capacity of the blowers in operation
AND the standby blower is available

THEN the blower(s) must be switched into operation as follows:

- Ramp down running blower(s) to an adjustable capacity C2
- Start the (additional) blower(s) with capacity C2
- When the blowers are in operation for an adjustable time, control them on the requested air capacity (working as a single unit).

Blower switching off

IF

- Multiple blowers are in operation
- AND the requested air capacity is lower than the total minimum capacity of the blowers in operation
- AND the blowers are running at a capacity below an adjustable value C3 for an adjustable time,

THEN the (supernumerary) blowers(s) must be switched off:

- Select the longest running blower(s) and stop it (them).
- Control the other blower(s) on the requested capacity.

IF

- No aeration is required

THEN

- Stop running blower(s).

4.4 Settings

- Settings for blowers and control
- Insufficient air capacity settings
- Blower capacity scaling settings

4.5 Interlocks / alarms

- Insufficient air capacity (for each reactor) *(available blower capacity lower than demand)*

- No blower available for Reactor # (for each reactor) (*alarm*)

4.6 Presentation

- Blowers in current configuration for each header.

4.7 Specification sheets

4.7.1 Blower MOD NVPP 001 bl 001... MOD NVPP 001 bl 006

Settings

SCADA

- Manual-0-Auto
- Start/stop
- Capacity setpoint (manual)
- Controller settings (gain, integration time)

Automation

Automatic

In automatic mode, the blowers are controlled as described in section 4.2.

Manual

In manual mode, the operator can start and stop the blower. The blower must be operated at a capacity setpoint, adjustable by the operator.

Interlocks

- Thermal overload (TS)
- Electric circuit failure
- Safety switch off
- Frequency inverter failure

Presentation

- State (automatic/ manual/ field control/ safety switch)
- Started (read back start is true)
- Stopped (read back stop is true)
- Failure
- Actual blower capacity

5 SLUDGE BUFFER

5.1 Introduction

The sludge of the sludge buffer is pumped to RWZI Weert. For the sludge buffer, 1 frequency-controlled pump is installed for sludge discharge.

5.2 Automation

Sludge pump flowcontrol

Valve MOD NVPP 001 a--029 should be opened followed by the sludge pump must being started at the H level setpoint of the sludge buffer. At the L level setpoint the sludge pump should be stopped followed by closing valve MOD NVPP 001 a-- 029. Macerator MOD NVPP 001 vsn 001 is in operation simultaneously with the sludge discharge pump. The pump must be controlled on the adjustable flow setpoint configured in SCADA.

5.3 Specification sheets

5.3.1 Sludge discharge pump MOD NVPP 001 sbp 001

Settings

SCADA

- Manual-0-Auto
- Start/Stop
- Capacity setpoint (Manual)
- Controller settings (gain, integration time)
- Settings Low level for dry run protection

Automation

Automatic

In automatic mode, the pump is controlled as described in section 5.2.

Manual

In manual mode, the operator can start and stop the pump. The pump must be operated at a capacity setpoint, adjustable by the operator.

Interlocks

- Thermal overload motor
- Electric circuit failure
- Safety switch off
- Frequency inverter failure

Presentation

- State (automatic/ manual/ field control/ safety switch)
- Started (read back start is true)

- Stopped (read back stop is true)
- Failure
- Actual pump capacity

5.3.2 Valve Nereda® sludge buffer MOD NVPP 001 a-- 029

Settings

SCADA

- Manual-0-Auto
- Open / Close

Automation

Automatic

In automatic mode, the valves must be operated as described in the various equipment phases under section 5.2.

Manual

In manual mode the operator can issue the following commands:

- ☐ Open
- ☐ Close

Interlocks

- Check on feedback failure (valve opening / closing)
- Electric circuit failure

Presentation

- State (automatic/ manual/ field control)
- Opened (read back open is true)
- Closed (read back close is true)
- Failure

5.3.3 Level measurement sludge buffer MOD NVPP 001 It 003

SCADA

- Level [m]

Switch points	Action	Ad	Pv	Al	Alarm text
HH	Abort / interlock sludge discharge phase -> operation conditions not met.	x		x	Level HH sludge buffer
H	Start sludge discharge pump Stop Sludge buffer flushing	x		x	Level High sludge buffer
L	Stop sludge discharge pump	x		x	Level Low sludge buffer

Failure measurement	Abort / interlock Feed phase sludge buffer -> operation conditions not met			x	Failure level measurement sludge buffer
---------------------	--	--	--	---	---

Ad = adjustable; Pv = Preference Value; Al = Alarm.

6 EFFLUENT

6.1 Introduction

The effluent from the Nereda® reactors is collected in the effluent gutter and drained to the effluent buffer tank. Effluent is pumped to RWZI Weert or pumped to the sludge buffer for flushing.

6.2 Automation

Effluent pump flowcontrol

The effluent pump must be started at the H level setpoint of the effluent buffer and stopped at the L level setpoint, or when a flush of the sludge buffer or grit screen installation is started. When the effluent pump is active (while no flushing is active), valve MOD NVPP 001 a—014 and --011 must be opened. If a flush is active, valve MOD NVPP 001 a—014 must be closed, the valves that must be opened are described in the following paragraphs. The pump must be controlled on the adjustable flow setpoint configured in SCADA. When the effluent pump is running and a switch of valves needs to occur, the pump first needs to go to minimum speed. When the pump is running at the minimum capacity, the valve positions can be switched. When the valves have switched their position, the pump must be controlled again on the adjustable flow setpoint configured in SCADA.

Flushing sludge buffer

Periodically the sludge buffer must be flushed using effluent. Flushing must be started based on an adjustable level setpoint in the sludge buffer or started in SCADA. The flushing of the sludge buffer will run for an adjustable time, or until the H level setpoint in the sludge buffer has been reached. When the effluent pump is active for flushing, valve MOD NVPP 001 a--017 must be opened. The pump must be controlled on the adjustable flow setpoint configured in SCADA.

Flushing grit screens

While in operation, the grit screens must be cyclically flushed with effluent. The duration of the flush cycle, and the duration of the flushing must be adjustable in SCADA. When the effluent pump is active for flushing the grit screens, valve MOD NVPP 001 a—012 must be opened. The pump must be controlled on the adjustable flow setpoint configured in SCADA.

Flushing grit screen installation

The grit screen installation must also be flushed using effluent. Flushing must be started when the grit screen installation (and feed pump) is turned off, and an adjustable maximum time since the last flush has been exceeded. The flush can also be started in SCADA when the screen installation is not in operation. The flushing of the screen installation will run for an adjustable time. When the effluent pump is active for flushing the grit screen installation, valves MOD NVPP 001 a—016 **AND** --002 must be opened. The pump must be controlled on the adjustable flow setpoint configured in SCADA.

Calamity pit

The calamity pit pump (46 2 AP Izp 95 001-002) must be operated must be started at the H level setpoint of the calamity pit and stopped at the L level setpoint, using level measurement 46 2 AP LT 95 001. Level

measurement 46 2 AP LT 95 001 must also have a H1-level setpoint (above the H-level setpoint) that generates an alarm indicating to the operator that the level in the calamity pit is too high. As a back-up to prevent overflowing of the calamity pit, level switch 46 2 AP LS 95 001 (HH-level) will be installed. When an HH-level of the level switch is detected, the installation should be stopped. This must be done by stopping (or closing) the following equipment and making it unavailable until the HH-level is no longer active:

- Influent feed pump 46 2 AP ifp 95 001
- Reactor feed pump MOD NVPP 001 ctp 001
- Sludge discharge pump MOD NVPP 001 sbp 001
- Effluent pump MOD NVPP 001 eff 001
- Valve Nereda® sludge buffer MOD NVPP 001 a-- 029
- Valves effluent MOD NVPP 001 a--14/--17

Flush to calamity pit

When the ambient temperature is below the freeze point (measured with MOD NVPP 001 the 002), the pipes towards the drainage well of RWZI Weert should be drained after use. After the sludge- effluent or calamity pump has been in operation, the corresponding drain valve should be opened to the calamity pit for an adjustable time.

Autosampler control

The autosampler must be controlled by the PLC by sending a pulse to the autosampler to start each sample. Samples must be taken iteratively (based on volume) when effluent is flowing out of the reactors. Samples must be taken with an adjustable time delay after the influent pump has been started to ensure that the level in the reactor has reached the V-notch level.

When the pump to the reactors is started an adjustable timer must be started. Starting the timer can be based on an adjustable flow setpoint or based on the start command of the influent pump MOD NVPP 001 ctp 001. Once the timer has finished (~5 mins adjustable time) The software should start tracking the volume of influent pumped and should send a pulse to the autosampler every time 1 m3 (adjustable volume) has been pumped.

Aurea installation

The feeding of the Aurea installation happens from the effluent buffer. The start and stop level of the effluent buffer to control the pumping must be configurable in SCADA. The flow setpoint at which the pump runs must also be a setting in SCADA. Since there is an overflow on the Aurea buffer, there is no need to interlock the feeding on high level.

When the level in the effluent buffer is above the start level, the pumping must start. This happens by opening the valve MOD NVPP 001 a—018 and after that running pump MOD NVPP 001 ctp 003 at the desired flow setpoint. For this flow measurement MOD NVPP 001 ft 003 is used.

When the level in the effluent buffer is below the stop level, the pumping must stop. This happens by stopping pump MOD NVPP 001 ctp 003 and after that closing valve MOD NVPP 001 a—018.

6.3 Settings

Adjustable values, as mentioned in the automation section above, must be made adjustable on SCADA.

6.4 Alarms

- None

6.5 Presentation

- None

6.6 Specification sheets

6.6.1 Effluent pump MOD NVPP 001 eff 001

Settings

SCADA

- Manual-0-Auto
- Start/Stop
- Capacity setpoint (Manual)
- Controller settings (gain, integration time)
- Settings Low level for dry run protection

Automation

Automatic

In automatic mode, the pumps are controlled as described in section 5.2.

Manual

In manual mode, the operator can start and stop the pump. The pump must be operated at a capacity setpoint, adjustable by the operator.

Interlocks

- Thermal overload motor
- Electric circuit failure
- Safety switch off
- Frequency inverter failure
- LL-flow (time delayed)

Presentation

- State (automatic/ manual/ field control/ safety switch)
- Started (read back start is true)
- Stopped (read back stop is true)
- Failure
- Actual pump capacity

6.6.2 Valves effluent MOD NVPP 001 a--14/--17

Settings

SCADA

- Manual-0-Auto

- Open / Close

Automation

Automatic

In automatic mode, the valves must be operated as described in the various equipment phases under section 5.2.

Manual

In manual mode the operator can issue the following commands:

- Open
- Close

Interlocks

- Check on feedback failure (valve opening / closing)
- Electric circuit failure

Presentation

- State (automatic/ manual/ field control)
- Opened (read back open is true)
- Closed (read back close is true)
- Failure

6.6.3 Turbidity/dry solids measurement effluent MOD NVPP 001 40QT2101

SCADA

- Dry solids [mg/l]

Switch points	Action	Ad	Pv	Al	Alarm text
H		x		x	Dry solids High effluent
Failure measurement				x	Failure dry solids measurement effluent

Ad = adjustable; Pv = Preference Value; Al = Alarm.

6.6.4 Phosphate PO₄ measurement effluent MOD NVPP 001 40QT2102

SCADA

- Phosphate concentration [mg/l]

Switch points	Action	Ad	Pv	Al	Alarm text
Failure measurement				x	Failure phosphate measurement effluent

Ad = adjustable; Pv = Preference Value; Al = Alarm.

6.6.5 Ammonium NH₄ measurement MOD NVPP 001 40QT2104

SCADA

- Ammonium concentration [mg/l]

Switch points	Action	Ad	Pv	Al	Alarm text
Failure measurement	Action is taken by Nereda Controller, no action from PLC required. See "Section 4 - Process automation and ICT"			x	Failure ammonium measurement effluent

Ad = adjustable; Pv = Preference Value; Al = Alarm.

6.6.6 Nitrate NO₃ measurement MOD NVPP 001 40QT2105

SCADA

- Nitrate concentration [mg/l]

Switch points	Action	Ad	Pv	Al	Alarm text
Failure measurement	Action is taken by Nereda Controller, no action from PLC required. See "Section 4 - Process automation and ICT"			x	Failure nitrate measurement effluent

Ad = adjustable; Pv = Preference Value; Al = Alarm.

6.6.7 Auto sampler MOD NVPP 001 40QT2103

SCADA

Switch points	Action	Ad	Pv	Al	Alarm text
Failure measurement				x	Failure auto sampler effluent

Ad = adjustable; Pv = Preference Value; Al = Alarm.

6.6.8 Flow measurement effluent MOD NVPP 001 ft 002

SCADA

- Flow [m³/h]

Switch points	Action	Ad	Pv	Al	Alarm text
HH		x		x	Flow too high
L		x		x	Flow low
LL	Stop effluent pump	x		x	Flow too low (time delayed)
Deviation alarm		x		x	Deviation alarm feed flow
Failure measurement				x	Failure flow measurement

Ad = adjustable; Pv = Preference Value; Al = Alarm.

6.6.9 Level measurement calamity pit 46 2 AP LT 95 001

SCADA

■ Level [m]

Switch points	Action	Ad	Pv	Al	Alarm text
H1		x		x	Level too High calamity pit
H	Start calamity pit pump	x			
L	Stop calamity pit pump	x			
Failure measurement	Abort / interlock Feed, Aerate phase -> operation conditions not met			x	Failure level measurement calamity pit

Ad = adjustable; Pv = Preference Value; Al = Alarm.

6.6.10 Level switch calamity pit 46 2 AP LS 95 001

SCADA

■ Level switch point

Switch points	Action	Ad	Pv	Al	Alarm text
HH	Stop the installation (See 6.2 Calamity pit)			x	Level too High calamity pit
Failure measurement				x	Failure level switch calamity pit

Ad = adjustable; Pv = Preference Value; Al = Alarm.

7 EMERGENCY RECIPE

7.1 Introduction

As described in the Control Philosophy, an Emergency Recipe is a safeguard located in and controlled by the PLC that is to be used in exceptional cases. This comprises of a series of equipment phases which are carried out sequentially on phase time basis.

7.2 Configuration

The commands that are sent to the phases by the Emergency Recipe have to be handled in the same way as the commands given from the Nereda® Controller.

In the Emergency Recipe only one phase can be active at a time. The next phase may only be started when the previous one is completed (READY / INTERLOCKED).

7.2.1 Emergency scheduler reactors

An Emergency Recipe for a reactor must be started when:

- The operation mode for the Nereda® Controller is switched via the process automation to 'Emergency' (automatically in case of communication failure or manually by operator),
- AND the Emergency Recipe for the reactor is set to 'On' (only possible if no phase Aborted).

When the operation mode of the Nereda controller is switched between different process control modes (see also appendix S4.1 Control Philosophy), all phases must first receive a STOP command in case of 'discontinuous feed'. With 'continuous feed' all phases must receive the STOP command except the 'feed' phase, which must remain active.

Before the Emergency Recipes for the Nereda® reactors are started for the first time, an adjustable start-up delay time (minutes) must have elapsed. After expiration of the delay time, all the reactors are in an order determined by the previous feed phases. The Nereda Controller also takes into account the actual feed-status of the reactors during emergency mode while scheduling the first reactor to feed after switching from 'emergency' to 'auto' mode.

Therefore, the PLC must keep track of the last feed phase of each reactor (MinutesSinceLastFeed signal). Based on this time it will determine a reactor number for the first, second, third reactor to start the Emergency Recipe). The reactor fed the longest time ago has to start as the first reactor.

The first reactor of a treatment line starts its Emergency Recipe with the feed phase and a cycle time of $CT_{reactor\ actual}$. The second reactor (of the same line) starts feeding $CT_{reactor\ actual} / N_{available\ reactors}$ minutes after the previous reactor has started feeding.

Before feeding, the second reactor starts its emergency recipe in step 3. It starts at the same time as the first reactor and follows the regular Emergency Recipe steps according to the procedure prescribed in **Table 3**. Since the second reactor should be ready to feed after $CT_{reactor\ actual} / N_{available\ reactors}$ minutes, the aerate time will be reduced to:

$$T_{step\ 5} = \frac{CT_{reactor\ actual}}{N_{available\ reactors}} - T_{step\ 3} - T_{step\ 4} - T_{step\ 6} - T_{step\ 7} - T_{step\ 8} \quad \text{minutes}$$

If the calculated aeration time is less than 15 minutes, the reactor should not start but wait for its 'feed' time to become active.

Consequently, reactor i in a treatment line of more than two reactors also starts its emergency recipe in step 3. However, it has to wait for $(i - 2) \times T_{step\ 3}$ minutes before starting, since only 1 reactor can be in step 3 at the same time. Since reactor i should be ready to feed after $(i - 1) \times CT_{reactor\ actual} / N_{available\ reactors}$ minutes, the aerate time will be adjusted to:

$$T_{step\ 5} = \frac{(i-1) \times CT_{reactor\ actual}}{N_{available\ reactors}} - (i - 1) \times T_{step\ 3} - T_{step\ 4} - T_{step\ 6} - T_{step\ 7} - T_{step\ 8} \quad \text{minutes}$$

By way of illustration, the diagrams below show the Emergency Recipe scheduling for a 4-reactor continuous feed and a 3-reactor discontinuous feed configuration.

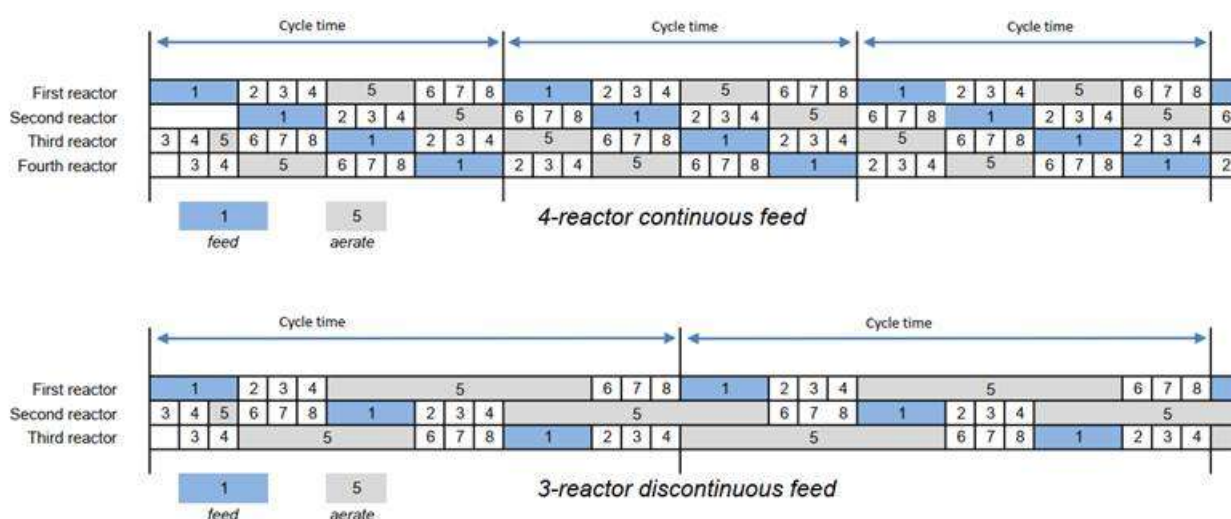


Figure 3 – Emergency Recipe scheduling 4-reactor continuous feed and 3-reactor discontinuous feed

Note: if there are two lines, the same procedure described above is applied to line 2. The first reactor of line 2 starts feeding $CT_{reactor\ actual} / (2 \times N_{available\ reactors})$ after the first reactor of line 1 has started feeding.

If during the Emergency Recipe an active phase, with the exception of the phases Sludge discharge and Lower level, reaches the Aborted state, or if the phase which has to be started is not ready, the corresponding reactor must be stopped and switched out of emergency control mode by the PLC (aborting the Emergency Recipe for this reactor, i.e. emergency mode set to 'Off'). After solving the problem, the operator can reset the phase, he can also switch the reactor back into emergency mode.

Reactor cycle time

The reactors must be operated with a pre-set adjustable nominal cycle time which is the same for all reactors. The actual cycle time ($CT_{reactor\ actual}$) for a reactor depends on the number of available reactors:

$$CT_{reactor\ actual} = \frac{N_{available\ reactors}}{N_{total\ reactors}} \times CT_{nominal} \quad [\text{min}]$$

With $N_{available\ reactors}$ = number of reactors with emergency mode set to 'on' (with $N_{available\ reactors} \geq 2$)

To synchronize the reactors mutually, the next available reactor starts feeding $CT_{reactor\ actual} / N_{available\ reactors}$ minutes after the previous reactor started feeding.

Influent buffer

With an influent buffer available, the phase times for each phase must be calculated as follows:

- All phases except *aerate*: fixed, adjustable time.
- *Aerate*: cycle time minus phase times of all other phases.

When the number of available reactors changes during operation, the interval time must be recalculated immediately. With $N-1$ ($N \geq 2$) available reactors, the actual cycle time of remaining available reactors is thereby shortened, the actual phase time of all equipment phases except phase 'aerate' remain the same.

7.2.2 Emergency Recipe Nereda® reactor

An Emergency Recipe for the Nereda® reactors can consist of the following steps (*values to be determined during start-up*):

Table 3 - Emergency Recipe Nereda® reactors

Step	Active phase	Duration / capacity (adjustable via SCADA)
1	Feed	0....33 m3/h 0....60 minutes (OR stop on Low Level influent buffer)
2	Wait	0....60 minutes
3	Sludge discharge**	0....30 minutes (OR stop on Reactor level $\leq$ L1)
4	Lower level	0 minutes (<i>not applicable</i>) (OR stop on Reactor level $\leq$ L1)
5	Aerate	0....100% aeration capacity setpoint (variable calculated phase time*) (Default: Aeration grid)
6	Vent	0....15 minutes (parameter selection: sludge grid [0/1], air grid [0/1], influent grid [0/1] adjustable; default: all)
7	Sludge discharge**	0....30 minutes (OR stop on Reactor level $\leq$ L2)
8	Wait	0....60 minutes
		<i>Pre-set nominal Cycle time $CT_{nominal}$: 30....360 minutes</i>
		*) <i>Calculated Aeration phase time = Cycle time - SUM (phase times 1-4, 6-8)</i> **) <i>If the recipe has had a sludge discharge in step 3, it should skip the sludge discharge in step 7. Depending on the choice, 0 min for the phase time must be entered in step 3 or in step 7.</i>

During the feed phase, once the level in the influent buffer has reached an adjustable Low level, or when an adjustable feed time has elapsed, the feed phase of the reactor must stop.

Before aeration, the level in the reactor must be lowered to an adjustable value L1.

The phase time of all equipment phases in the recipe (as is displayed in **Table 3**) must be adjustable per step, except for step 5: Aeration phase. The phase time for the aeration phase must be calculated as the nominal cycle time minus the sum of the other phases. Note that this must be a continuous calculation as the feed *phase* and the sludge discharge phase do not stop solely based on time.

If one or more of the phases before aeration stop earlier (on level), the remaining phase time is added to the aeration time in step 5. The remaining phase time of phases after aeration (sludge discharge) is added to the waiting time in step 8.

7.3 Settings

On SCADA the operator must be able to adjust the following parameters for the Emergency Recipe:

- Nereda® Emergency Recipe start-up delay time
- Nereda® reactor cycle time

For the Nereda® reactors:

- See table in section 7.2.2.

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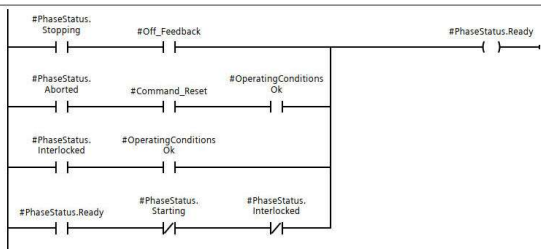
AquasuiteControlPhase_LAD [FB3]

AquasuiteControlPhase_LAD Properties

General							
Name	AquasuiteControl-Phase_LAD	Number	3	Type	FB	Language	LAD
Numbering	automatic						
Information							
Title	Aquasuite Control Phase	Author		Comment		Family	
Version	0.1	User-defined ID					

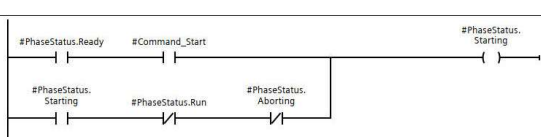
Name	Data type	Default value	Retain	Accessible from HMI	Visible in HMI	Setpoint	Comment
Input							
OperatingConditionsOk	Bool	false	Non-retain	False	False	False	
On_Feedback	Bool	false	Non-retain	False	False	False	
Off_Feedback	Bool	false	Non-retain	False	False	False	
Command_Start	Bool	false	Non-retain	False	False	False	
Command_Stop	Bool	false	Non-retain	False	False	False	
Command_Reset	Bool	false	Non-retain	False	False	False	
Output							
Ready	Bool	false	Non-retain	False	False	False	
Starting	Bool	false	Non-retain	False	False	False	
Run	Bool	false	Non-retain	False	False	False	
Stopping	Bool	false	Non-retain	False	False	False	
Aborting	Bool	false	Non-retain	False	False	False	
Aborted	Bool	false	Non-retain	False	False	False	
Interlocked	Bool	false	Non-retain	False	False	False	
InOut							
Static							
PhaseStatus	Struct		Retain	False	False	False	
Ready	Bool	false	Retain	False	False	False	
Starting	Bool	false	Retain	False	False	False	
Run	Bool	false	Retain	False	False	False	
Stopping	Bool	false	Retain	False	False	False	
Aborting	Bool	false	Retain	False	False	False	
Aborted	Bool	false	Retain	False	False	False	
Interlocked	Bool	false	Retain	False	False	False	
Temp							
Constant							

Network 1: Ready



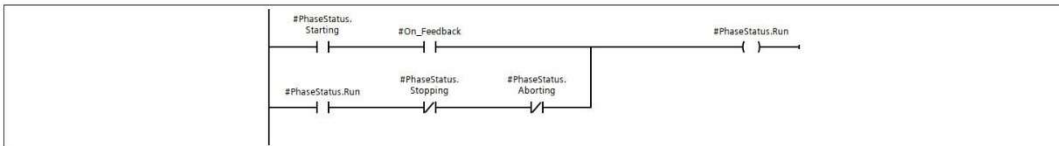
Symbol	Address	Type	Comment
#Command_Reset		Bool	
#Off_Feedback		Bool	
#OperatingConditionsOk		Bool	
#PhaseStatus.Aborted		Bool	
#PhaseStatus.Interlocked		Bool	
#PhaseStatus.Ready		Bool	
#PhaseStatus.Starting		Bool	
#PhaseStatus.Stopping		Bool	

Network 2: Starting



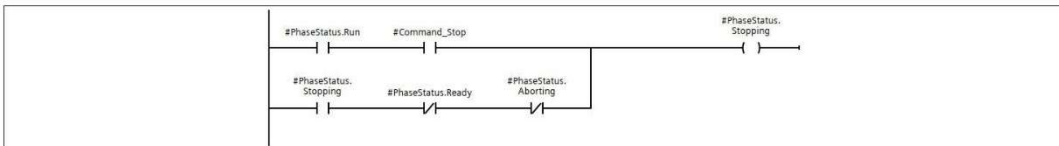
Totally Integrated Automation Portal			
Symbol	Address	Type	Comment
#Command_Start		Bool	
#PhaseStatus.Aborting		Bool	
#PhaseStatus.Ready		Bool	
#PhaseStatus.Run		Bool	
#PhaseStatus.Starting		Bool	

Network 3: Run



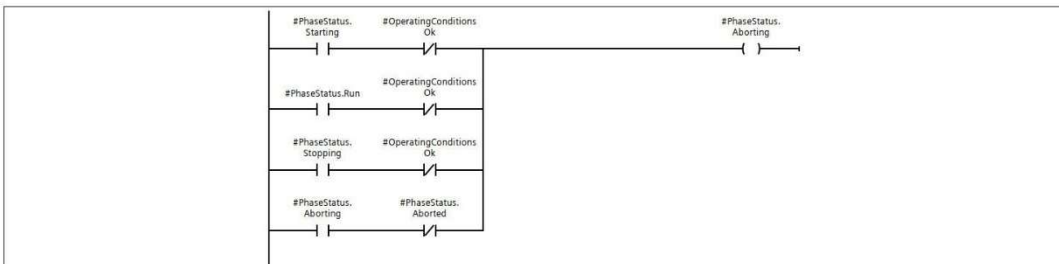
Symbol	Address	Type	Comment
#On_Feedback		Bool	
#PhaseStatus.Aborting		Bool	
#PhaseStatus.Run		Bool	
#PhaseStatus.Starting		Bool	
#PhaseStatus.Stopping		Bool	

Network 4: Stopping



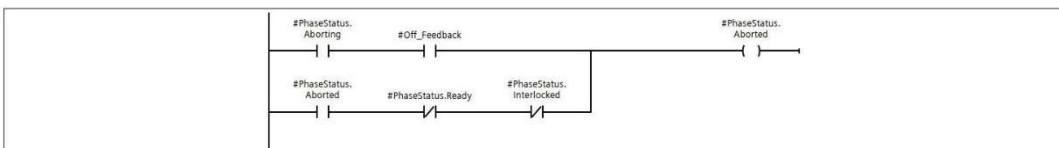
Symbol	Address	Type	Comment
#Command_Stop		Bool	
#PhaseStatus.Aborting		Bool	
#PhaseStatus.Ready		Bool	
#PhaseStatus.Run		Bool	
#PhaseStatus.Stopping		Bool	

Network 5: Aborting

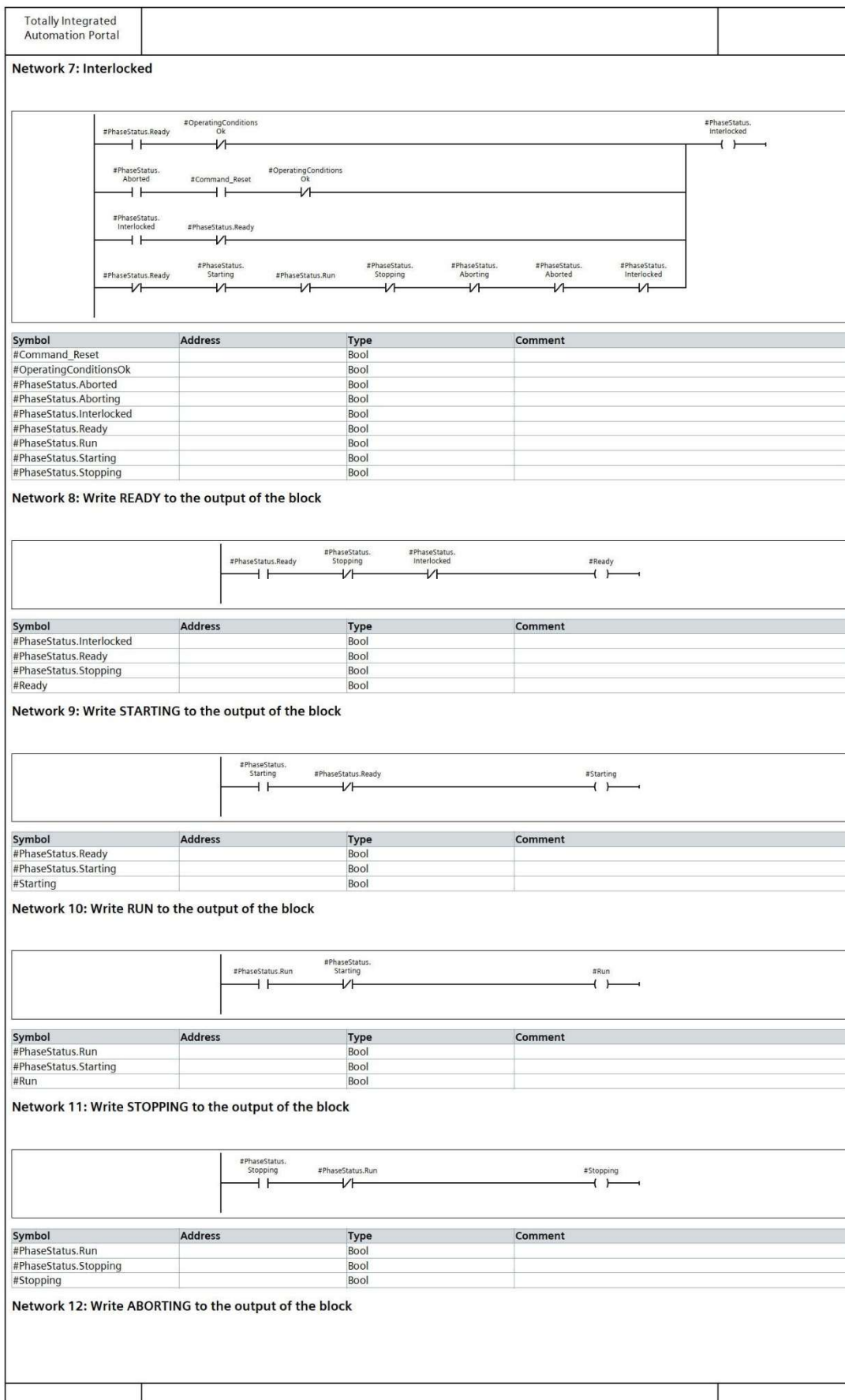


Symbol	Address	Type	Comment
#OperatingConditionsOk		Bool	
#PhaseStatus.Aborted		Bool	
#PhaseStatus.Aborting		Bool	
#PhaseStatus.Run		Bool	
#PhaseStatus.Starting		Bool	
#PhaseStatus.Stopping		Bool	

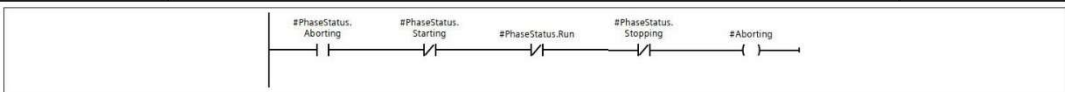
Network 6: Aborted



Symbol	Address	Type	Comment
#Off_Feedback		Bool	
#PhaseStatus.Aborted		Bool	
#PhaseStatus.Aborting		Bool	
#PhaseStatus.Interlocked		Bool	
#PhaseStatus.Ready		Bool	



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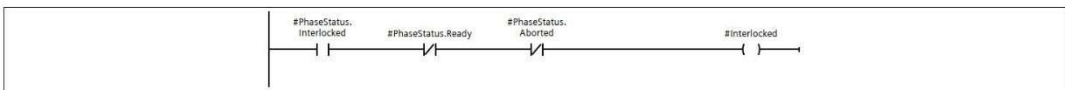
Symbol	Address	Type	Comment
#Aborting		Bool	
#PhaseStatus.Aborting		Bool	
#PhaseStatus.Run		Bool	
#PhaseStatus.Starting		Bool	
#PhaseStatus.Stopping		Bool	

Network 13: Write ABORTED to the output of the block



Symbol	Address	Type	Comment
#Aborted		Bool	
#PhaseStatus.Aborted		Bool	
#PhaseStatus.Aborting		Bool	

Network 14: Write INTERLOCKED to the output of the block



Symbol	Address	Type	Comment
#Interlocked		Bool	
#PhaseStatus.Aborted		Bool	
#PhaseStatus.Interlocked		Bool	
#PhaseStatus.Ready		Bool	

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AquasuiteControlPhase_ST [FB2]							
AquasuiteControlPhase_ST Properties							
General							
Name	AquasuiteControlPhase_ST	Number	2	Type	FB		
Numbering	automatic			Language	SCL		
Information							
Title		Author		Comment			
Version	0.1	User-defined ID		Family			
Name	Data type	Default value	Retain	Accessible from HMI	Visible in HMI	Setpoint	Comment
▼ Input							
OperatingConditionsOK	Bool	false	Non-retain	False	False	False	
On_Feedback	Bool	false	Non-retain	False	False	False	
Off_Feedback	Bool	false	Non-retain	False	False	False	
Command_Start	Bool	false	Non-retain	False	False	False	
Command_Stop	Bool	false	Non-retain	False	False	False	
Command_Reset	Bool	false	Non-retain	False	False	False	
▼ Output							
Status	Int	0	Non-retain	False	False	False	
Ready	Bool	false	Non-retain	False	False	False	
Starting	Bool	false	Non-retain	False	False	False	
Run	Bool	false	Non-retain	False	False	False	
Stopping	Bool	false	Non-retain	False	False	False	
Aborting	Bool	false	Non-retain	False	False	False	
Aborted	Bool	false	Non-retain	False	False	False	
Interlocked	Bool	false	Non-retain	False	False	False	
InOut							
▼ Static							
PhaseStatus	Int	7	Retain	False	False	False	
Temp							
▼ Constant							
cReady	Int	0					
cStarting	Int	1					
cRun	Int	2					
cStopping	Int	3					
cAborting	Int	5					
cAborted	Int	6					
cInterlocked	Int	7					
<pre> 0001 #Ready := false; 0002 #Starting := false; 0003 #Run := false; 0004 #Stopping := false; 0005 #Aborting := false; 0006 #Aborted := false; 0007 #Interlocked := false; 0008 0009 CASE #PhaseStatus OF 0010 0: // Ready 0011 #Ready := true; 0012 0013 IF #Command_Start THEN 0014 #PhaseStatus := #cStarting; 0015 END_IF; 0016 0017 IF NOT #OperatingConditionsOK THEN 0018 #PhaseStatus := #cInterlocked; 0019 END_IF; 0020 1: // Starting 0021 #Starting := true; 0022 0023 IF #On_Feedback THEN 0024 #PhaseStatus := #cRun; 0025 END_IF; 0026 0027 IF NOT #OperatingConditionsOK THEN 0028 #PhaseStatus := #cAborting; 0029 END_IF; 0030 2: // Run 0031 #Run := true; 0032 0033 IF #Command_Stop THEN 0034 #PhaseStatus := #cStopping; 0035 END_IF; 0036 0037 IF NOT #OperatingConditionsOK THEN 0038 #PhaseStatus := #cAborting; 0039 END_IF; 0040 3: // Stopping 0041 #Stopping := true; 0042 0043 IF #Off_Feedback THEN </pre>							

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```
0044     #PhaseStatus := #cReady;
0045     END_IF;
0046
0047     IF NOT #OperatingConditionsOK THEN
0048         #PhaseStatus := #cAborting;
0049     END_IF;
0050 5:    // Aborting
0051     #Aborting := true;
0052
0053     IF #Off_Feedback THEN
0054         #PhaseStatus := #cAborted;
0055     END_IF;
0056 6:    // Aborted
0057     #Aborted := true;
0058
0059     IF #Command_Reset AND #OperatingConditionsOK THEN
0060         #PhaseStatus := #cReady;
0061     END_IF;
0062
0063     IF #Command_Reset AND NOT #OperatingConditionsOK THEN
0064         #PhaseStatus := #cInterlocked;
0065     END_IF;
0066 7:    // Interlocked
0067     #Interlocked := true;
0068
0069     IF #OperatingConditionsOK THEN
0070         #PhaseStatus := #cReady;
0071     END_IF;
0072 ELSE
0073     #PhaseStatus := #cInterlocked;
0074 END_CASE;
0075
0076 #Status := #PhaseStatus;
```

Symbol	Address	Type	Comment
#Aborted		Bool	
#Aborting		Bool	
#cAborted	6	Int	
#cAborting	5	Int	
#cInterlocked	7	Int	
#Command_Reset		Bool	
#Command_Start		Bool	
#Command_Stop		Bool	
#cReady	0	Int	
#cRun	2	Int	
#cStarting	1	Int	
#cStopping	3	Int	
#Interlocked		Bool	
#Off_Feedback		Bool	
#On_Feedback		Bool	
#OperatingConditionsOK		Bool	
#PhaseStatus		Int	
#Ready		Bool	
#Run		Bool	
#Starting		Bool	
#Status		Int	
#Stopping		Bool	